

Persamaan ruang keadaan untuk fungsi transfer.

- Bentuk kanonik keterkontrolan
- Bentuk kanonik keterobservasian
- Bentuk kanonik diagonal
- Bentuk kanonik Jordan
- Signal-flow-graph untuk persamaan ruang keadaan

Bentuk kanonik keterkontrolan

Perhatikan model sistem linear orde- n

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} u' + b_n u.$$

Persamaan diatas dapat juga ditulis dalam bentuk:

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

Sistem ini termasuk SISO.

Persamaan ruang keadaan dalam bentuk kanonik keterkontrolan dari system diatas dapat ditulis dalam bentuk:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [b_n - a_n b_0 \quad b_{n-1} - a_{n-1} b_0 \quad \cdots \quad b_1 - a_1 b_0] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

Langkah-langkah untuk menunjukkan persamaan tersebut adalah sbb:

Dengan membagi langsung persamaan sebelumnya dapat dirubah menjadi:

$$\frac{Y(s)}{U(s)} = b_0 + \frac{(b_1 - a_1 b_0)s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0)s + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

atau $Y(s) = b_0 U(s) + \hat{Y}(s)$

dengan $\hat{Y}(s) = \frac{(b_1 - a_1 b_0)s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0)s + (b_n - a_n b_0)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} U(s)$

atau
$$\frac{\hat{Y}(s)}{(b_1 - a_1 b_0)s^{n-1} + \dots + (b_{n-1} - a_{n-1} b_0)s + (b_n - a_n b_0)}$$
$$= \frac{U(s)}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} = Q(s)$$

Dari persamaan terakhir, diperoleh persamaan berikut:

$$s^n Q(s) = -a_1 s^{n-1} Q(s) - \cdots - a_{n-1} s Q(s) - a_n Q(s) + U(s)$$

$$\begin{aligned} \hat{Y}(s) &= (b_1 - a_1 b_0) s^{n-1} Q(s) + \cdots + (b_{n-1} - a_{n-1} b_0) s Q(s) \\ &\quad + (b_n - a_n b_0) Q(s) \end{aligned}$$

Dengan memisalkan:

$X_1(s) = Q(s)$ $X_2(s) = sQ(s)$ $\cdot$ $\cdot$ $\cdot$ $X_{n-1}(s) = s^{n-2}Q(s)$ $X_n(s) = s^{n-1}Q(s)$		$sX_1(s) = X_2(s)$ $sX_2(s) = X_3(s)$ $\cdot$ $\cdot$ $\cdot$ $sX_{n-1}(s) = X_n(s)$		$\dot{x}_1 = x_2$ $\dot{x}_2 = x_3$ $\cdot$ $\cdot$ $\cdot$ $\dot{x}_{n-1} = x_n$
--	--	---	--	--

Dengan demikian persamaan ruang keadaan dalam bentuk kanonik keterkontrolan diperoleh.

Contoh

Tentukan persamaan ruang keadaan dalam bentuk kanonik keterkontrolan untuk system dengan fungsi transfer $G(s) = \frac{s+1}{s^2+2s+3}$.

Latihan

Diberikan sistem dengan persamaan ruang keadaan:

$$x'(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [b_3 \quad b_2 \quad b_1] x(t)$$

Tentukan fungsi transfer dari sistem diatas.

Bentuk kanonik keterobservasian

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \cdot \\ \cdot \\ \cdot \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ \cdot & \cdot & & \cdot & \cdot \\ \cdot & \cdot & & \cdot & \cdot \\ \cdot & \cdot & & \cdot & \cdot \\ 0 & 0 & \cdots & 1 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} + \begin{bmatrix} b_n - a_n b_0 \\ b_{n-1} - a_{n-1} b_0 \\ \cdot \\ \cdot \\ \cdot \\ b_1 - a_1 b_0 \end{bmatrix} u$$

$$y = [0 \quad 0 \quad \cdots \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_{n-1} \\ x_n \end{bmatrix} + b_0 u$$

Langka-Langkah untuk menunjukkanya

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

dapat dituliskan dalam bentuk

$$\begin{aligned} & s^n[Y(s) - b_0 U(s)] + s^{n-1}[a_1 Y(s) - b_1 U(s)] + \cdots \\ & + s[a_{n-1} Y(s) - b_{n-1} U(s)] + a_n Y(s) - b_n U(s) = 0 \end{aligned}$$

atau

$$\begin{aligned} Y(s) &= b_0 U(s) + \frac{1}{s} [b_1 U(s) - a_1 Y(s)] + \cdots \\ &+ \frac{1}{s^{n-1}} [b_{n-1} U(s) - a_{n-1} Y(s)] + \frac{1}{s^n} [b_n U(s) - a_n Y(s)] \end{aligned}$$

Dengan mendefinisikan variable keadaan:

$$X_n(s) = \frac{1}{s} [b_1 U(s) - a_1 Y(s) + X_{n-1}(s)]$$

$$X_{n-1}(s) = \frac{1}{s} [b_2 U(s) - a_2 Y(s) + X_{n-2}(s)]$$

·
·
·

$$X_2(s) = \frac{1}{s} [b_{n-1} U(s) - a_{n-1} Y(s) + X_1(s)]$$

$$X_1(s) = \frac{1}{s} [b_n U(s) - a_n Y(s)]$$

sehingga



$$sX_n(s) = X_{n-1}(s) - a_1 X_n(s) + (b_1 - a_1 b_0) U(s)$$

$$sX_{n-1}(s) = X_{n-2}(s) - a_2 X_n(s) + (b_2 - a_2 b_0) U(s)$$

·
·
·

$$sX_2(s) = X_1(s) - a_{n-1} X_n(s) + (b_{n-1} - a_{n-1} b_0) U(s)$$

$$sX_1(s) = -a_n X_n(s) + (b_n - a_n b_0) U(s)$$

diperoleh

$$Y(s) = b_0 U(s) + X_n(s)$$

Melalui invers transformasi Laplace, diperoleh:

$$\dot{x}_1 = -a_n x_n + (b_n - a_n b_0)u$$

$$\dot{x}_2 = x_1 - a_{n-1} x_n + (b_{n-1} - a_{n-1} b_0)u$$

.

.

.

$$\dot{x}_{n-1} = x_{n-2} - a_2 x_n + (b_2 - a_2 b_0)u$$

$$\dot{x}_n = x_{n-1} - a_1 x_n + (b_1 - a_1 b_0)u$$

dan

$$y = x_n + b_0 u$$

Dengan demikian bentuk kanonik keterobservasian dapat diperoleh.

Contoh

Tentukan persamaan ruang keadaan dalam bentuk kanonik keterobservasian dari system dengan fungsi transfer $G(s) = \frac{s+3}{s^2+3s+2}$.

Sistem $x' = Ax + Bu, y = Cx + Du$.

Fungsi transfer $G(s) = C(sI - A)^{-1}B + D$.
$$(sI - A)^{-1} = \frac{Adj(sI - A)}{|sI - A|}$$

Bentuk kanonik diagonal

Dalam hal fungsi transfer mempunyai pole pole yang berbeda semua, maka fungsi transfer tersebut dapat ditulis dalam bentuk:

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{(s + p_1)(s + p_2) \dots (s + p_n)}$$
$$= b_0 + \frac{c_1}{s + p_1} + \frac{c_2}{s + p_2} + \dots + \frac{c_n}{s + p_n}$$

Dengan memisalkan variabel keadaan:

$$X_1(s) = \frac{1}{s + p_1} U(s)$$

$$X_2(s) = \frac{1}{s + p_2} U(s)$$

·
·
·

$$X_n(s) = \frac{1}{s + p_n} U(s)$$

Diperoleh persamaan ruang keadaan diagonal

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \cdot \\ \cdot \\ \cdot \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & & & & 0 \\ & -p_2 & & & \\ & & \cdot & & \\ & & & \cdot & \\ & & & & \cdot \\ 0 & & & & -p_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ \cdot \\ \cdot \\ \cdot \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix} + b_0 u$$

Contoh

Tentukan persamaan ruang keadaan dalam bentuk kanonik diagonal untuk system dengan fungsi transfer $G(s) = \frac{(s+1)(s+4)}{s(s+2)(s+3)}$.

Bentuk kanonik Jordan

Misalkan pole yang pertama berulang sebanyak tiga kali, sehingga fungsi transfernya dapat ditulis dalam bentuk:

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{(s + p_1)^3 (s + p_4)(s + p_5) \cdots (s + p_n)}$$

atau

$$\frac{Y(s)}{U(s)} = b_0 + \frac{c_1}{(s + p_1)^3} + \frac{c_2}{(s + p_1)^2} + \frac{c_3}{s + p_1} + \frac{c_4}{s + p_4} + \cdots + \frac{c_n}{s + p_n}$$

Dengan mendefinikan
variable keadaan

$$X_1(s) = \frac{1}{(s + p_1)^3} U(s)$$

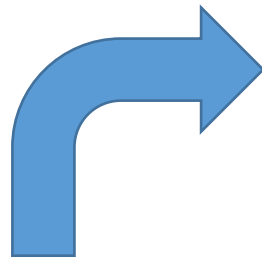
$$X_2(s) = \frac{1}{(s + p_1)^2} U(s)$$

$$X_3(s) = \frac{1}{s + p_1} U(s)$$

$$X_4(s) = \frac{1}{s + p_4} U(s)$$

⋮

$$X_n(s) = \frac{1}{s + p_n} U(s)$$



Diperoleh persamaan ruang keadaan
dalam bentuk Jordan

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & 1 & 0 & 0 & \cdots & 0 \\ 0 & -p_1 & 1 & \cdot & & \cdot \\ 0 & 0 & -p_1 & 0 & \cdots & 0 \\ \hline 0 & \cdots & 0 & -p_4 & & \\ \vdots & & \vdots & & \ddots & \\ \vdots & & \vdots & & & \ddots \\ 0 & \cdots & 0 & 0 & & -p_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u$$

$$y = [c_1 \quad c_2 \quad \cdots \quad c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

Contoh

Tentukan persamaan ruang keadaan dalam bentuk kanonik Jordan untuk system dengan fungsi transfer $G(s) = \frac{s(s+1)(s+4)}{(s+2)^3(s+3)}$.

SIMO Case: Let

1 Input : bentuk
kanonik
keterkontrolan

$$G(s) = \begin{pmatrix} g_1(s) \\ g_2(s) \\ \vdots \\ g_m(s) \end{pmatrix} = \frac{\beta_1 s^{n-1} + \beta_2 s^{n-2} + \dots + \beta_{n-1} s + \beta_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} + d,$$

where $\beta_i \in \mathbb{R}^m$ and $d \in \mathbb{R}^m$. Then

$$\begin{aligned} \dot{\vec{x}} &= A\vec{x} + bu \\ \vec{y} &= C\vec{x} + du \end{aligned}$$

$$G(s) = \left[\begin{array}{c|c} A & b \\ \hline C & d \end{array} \right], \quad b \in \mathbb{R}^n, \quad C \in \mathbb{R}^{m \times n}, \quad d \in \mathbb{R}^m$$

where

$$A := \begin{bmatrix} -a_1 & -a_2 & \dots & -a_{n-1} & -a_n \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix} \quad b := \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$
$$C = \begin{bmatrix} \beta_1 & \beta_2 & \dots & \beta_{n-1} & \beta_n \end{bmatrix}$$

$$\begin{aligned}
 Y(s) &= \beta_1 s^{n-1} X_n(s) + \beta_2 s^{n-2} X_n(s) + \dots + \beta_n X_n(s) + d U(s) \\
 &= (\beta_1 s^{n-1} + \beta_2 s^{n-2} + \dots + \beta_n) X_n(s) + d U(s) \\
 &= \frac{\beta_1 s^{n-1} + \beta_2 s^{n-2} + \dots + \beta_n}{s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_{n-1} s + a_n} U(s) + d U(s)
 \end{aligned}$$

$$\frac{Y(s)}{U(s)} = \frac{\beta_1 s^{n-1} + \beta_2 s^{n-2} + \dots + \beta_n}{s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_{n-1} s + a_n} + d$$

CONTOH : Tentukan pers. ruang keadaan dari sistem SISO :

$$G(s) = \begin{bmatrix} \frac{s+2}{s^2+3s+2} \\ \frac{2s+3}{s^2+3s+2} \end{bmatrix}$$

Pandang $G(s)$ sebagai $\Rightarrow g_1(s) = \frac{s+2}{s^2+3s+2}$ dan $g_2(s) = \frac{2s+3}{s^2+3s+2}$

Kita memiliki : $a_1 = 3$ dan $a_2 = 2$

Terdapat dua output : $\beta_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$\beta_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Maka pers. ruang keadaan : $A = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $C = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ $d = 0$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$\vec{y} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Tentukan persamaan ruang keadaan dari sistem:

$$G(s) = \begin{bmatrix} \frac{s+2}{s^2+3s+2} \\ \frac{2s+3}{s^2+3s+2} \end{bmatrix}$$

MISO Case: Let

1 output : bentuk kanonik
keterobservasian

$$y = x_1$$

$$G(s) = (g_1(s) \quad g_2(s) \quad \dots \quad g_p(s))$$

$$= \frac{\eta_1 s^{n-1} + \eta_2 s^{n-2} + \dots + \eta_{n-1} s + \eta_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} + d$$

with $\eta_i^*, d^* \in \mathbb{R}^p$. Then

$$G(s) = \left[\begin{array}{ccccc|c} -a_1 & 1 & 0 & \dots & 0 & \eta_1 \\ -a_2 & 0 & 1 & \dots & 0 & \eta_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ -a_{n-1} & 0 & 0 & \dots & 1 & \eta_{n-1} \\ -a_n & 0 & 0 & \dots & 0 & \eta_n \\ \hline 1 & 0 & 0 & \dots & 0 & d \end{array} \right]$$

	SIMO	MISO
B	$[1 \ 0 \ 0 \ \dots \ 0]^T$	Koefisien pembilang η
C	Koefisien pembilang β	$[1 \ 0 \ 0 \ \dots \ 0]$

CONTOH : Diberikan sistem MISO :

$$G(s) = \frac{1}{s^3 + 4s^2 + 5s + 2} \begin{bmatrix} 2s^2 + 3s + 1 & s^2 + 4s + 3 \end{bmatrix} \text{ ada 2 input}$$

diperoleh : $a_1 = 4$ $a_2 = 5$ $a_3 = 2$

$$\eta_1 = [2 \ 1] \quad \eta_2 = [3 \ 4] \quad \eta_3 = [1 \ 3]$$

Pers. Ruang Keadaan :

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -4 & 1 & 0 \\ -5 & 0 & 1 \\ -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 3 & 4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$y = x_1 = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Contoh: Tentukan persamaan ruang keadaan dari sistem:

$$G(s) = \begin{bmatrix} \frac{s+2}{s^2+3s+2} & \frac{2s+3}{s^2+3s+2} \end{bmatrix}$$

Signal-Flow Graphs of State Equations

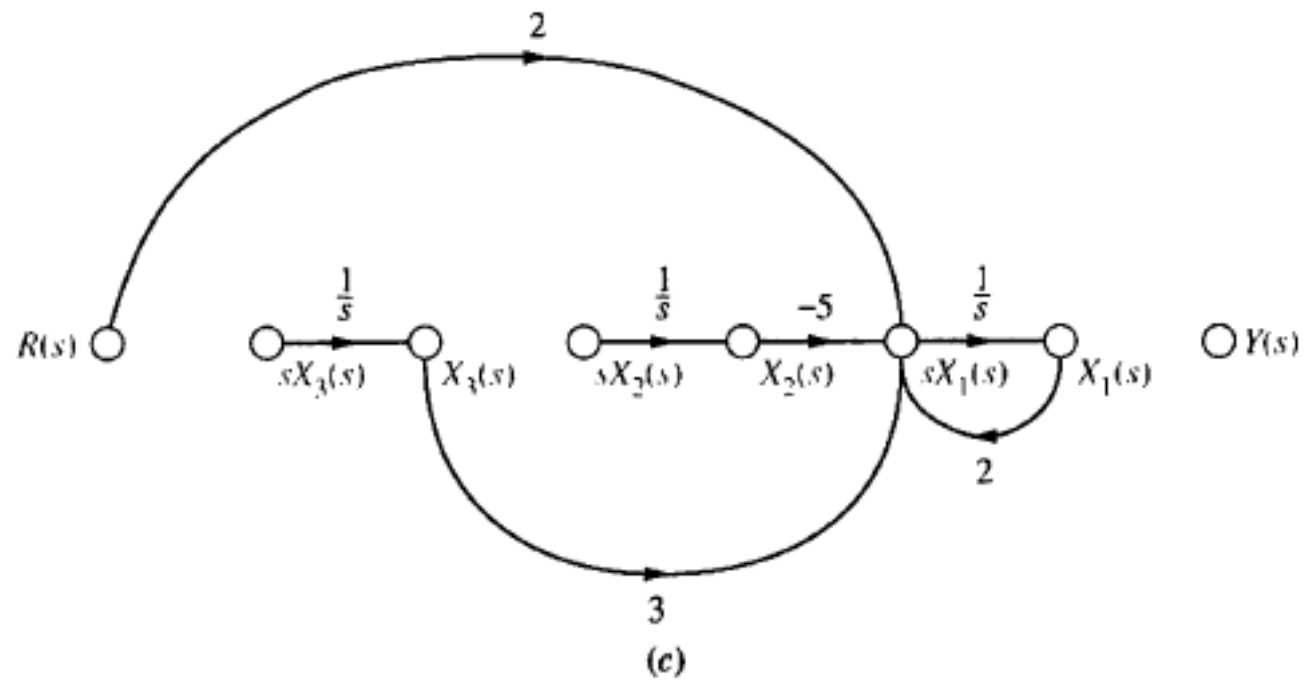
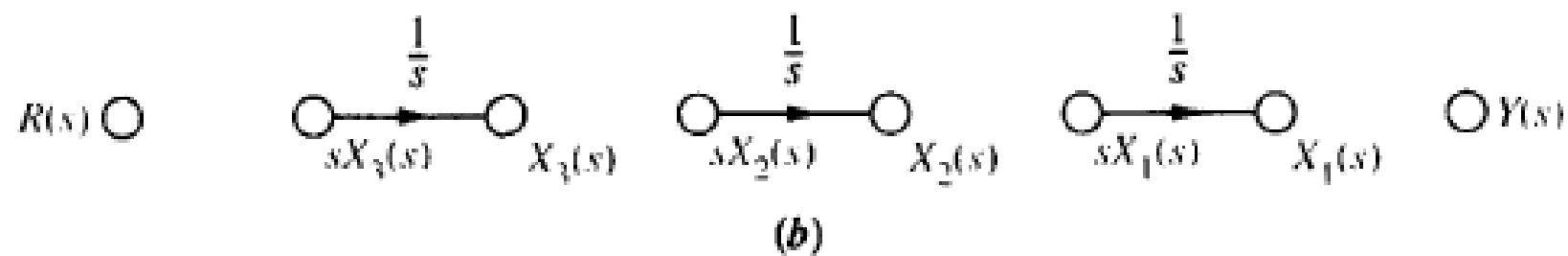
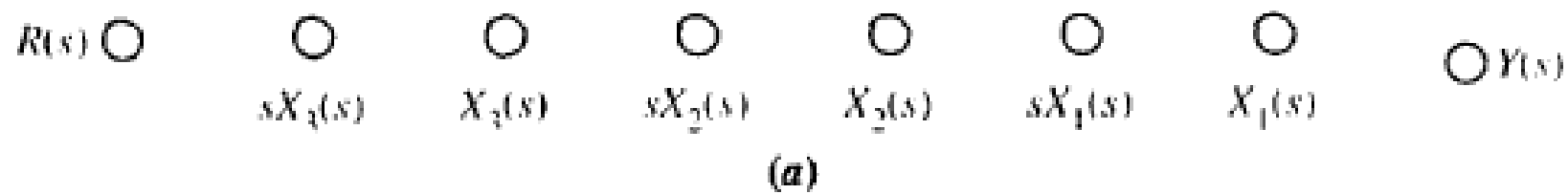
Consider the following state and output equations:

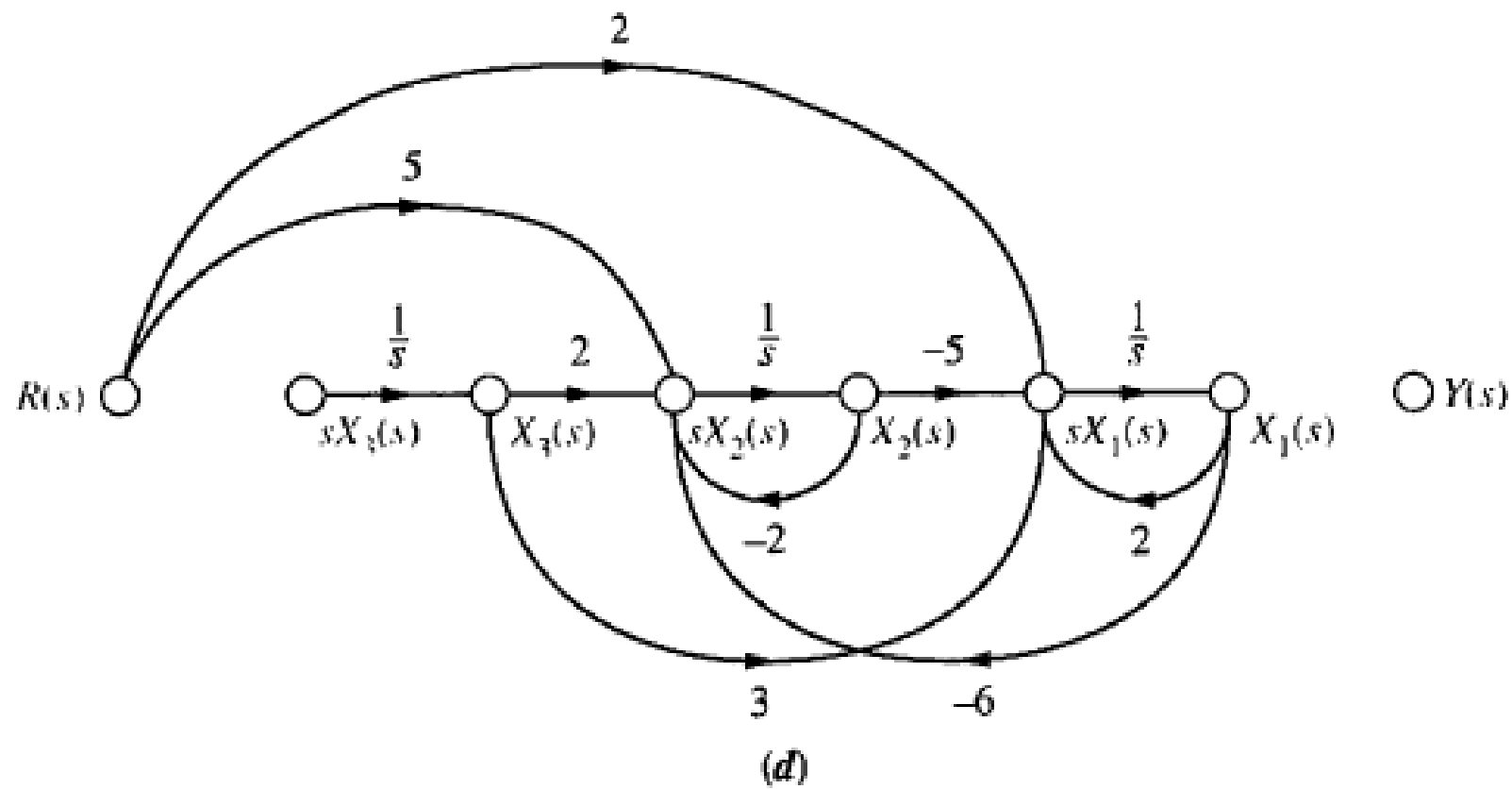
$$\dot{x}_1 = 2x_1 - 5x_2 + 3x_3 + 2r$$

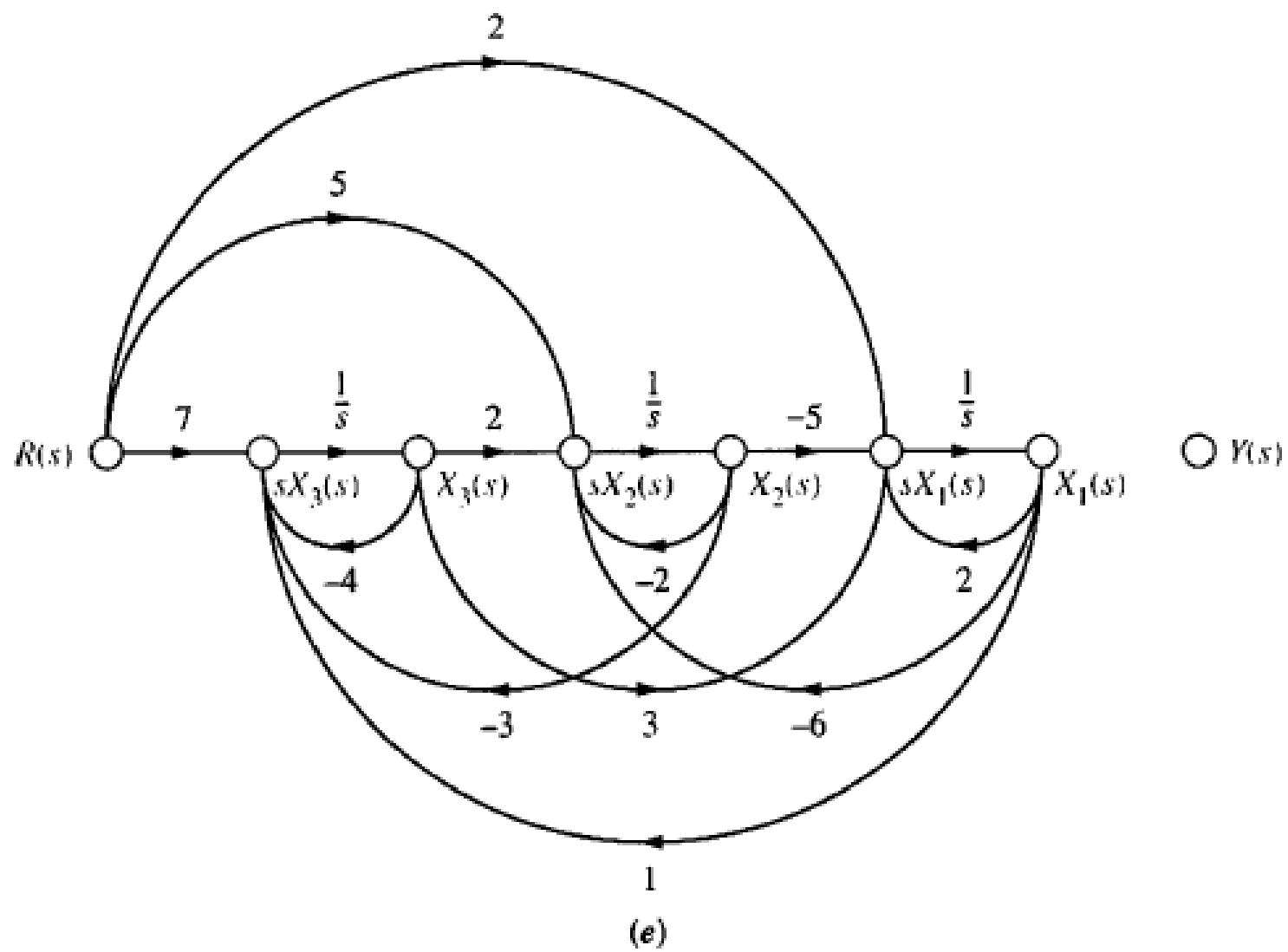
$$\dot{x}_2 = -6x_1 - 2x_2 + 2x_3 + 5r$$

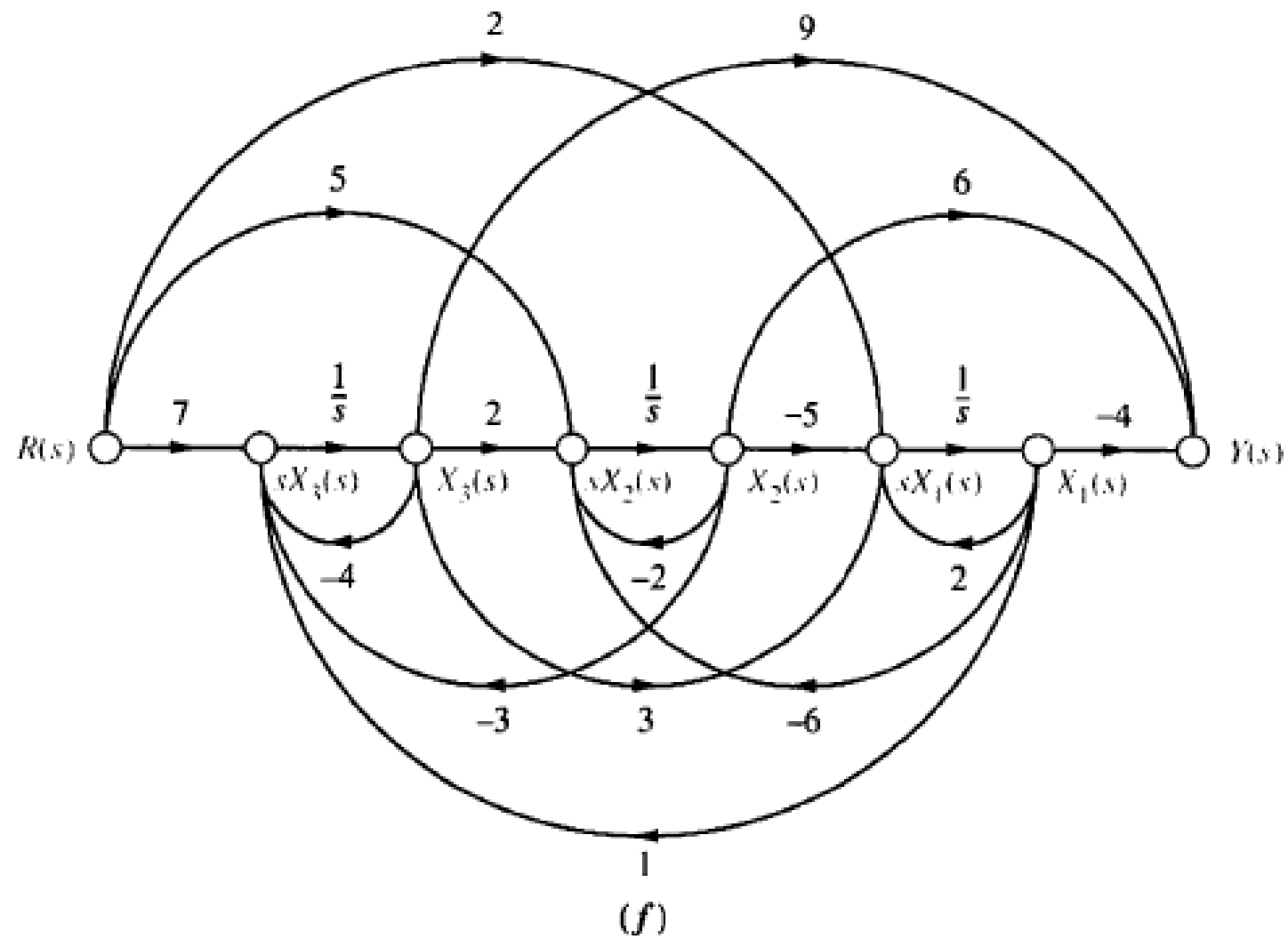
$$\dot{x}_3 = x_1 - 3x_2 - 4x_3 + 7r$$

$$y = -4x_1 + 6x_2 + 9x_3$$









Latihan

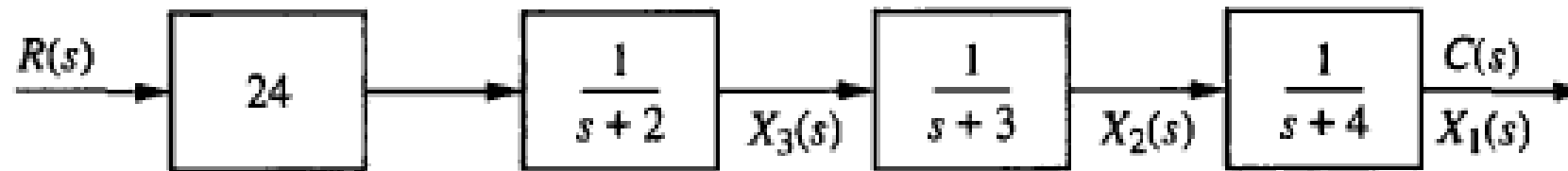
Draw a signal-flow graph for the following state and output equations:

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -3 & 1 \\ -3 & -4 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

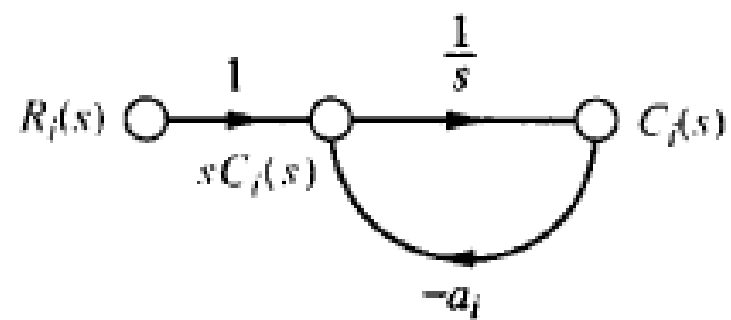
$$y = [0 \quad 1 \quad 0] \mathbf{x}$$

Cascade Form

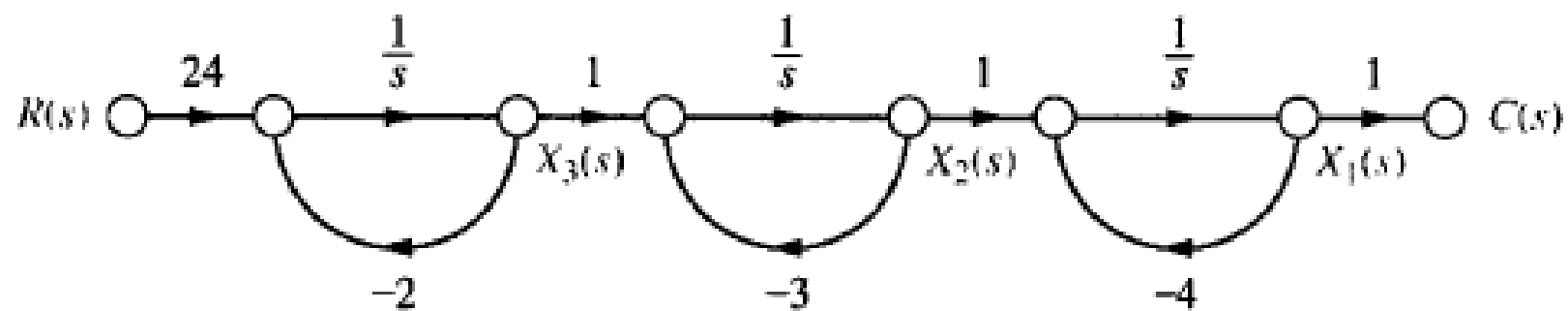
$$\frac{C(s)}{R(s)} = \frac{24}{(s+2)(s+3)(s+4)}$$



$$\frac{C_i(s)}{R_i(s)} = \frac{1}{(s + a_i)}$$



(a)



(b)

$$\dot{x}_1 = -4x_1 + x_2$$

$$\dot{x}_2 = \quad \quad -3x_2 + x_3$$

$$\dot{x}_3 = \quad \quad \quad -2x_3 + 24r$$

$$y = c(t) = x_1$$

$$\dot{\mathbf{x}} = \begin{bmatrix} -4 & 1 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

$$y = [1 \quad 0 \quad 0] \mathbf{x}$$

Parallel Form

$$\frac{C(s)}{R(s)} = \frac{24}{(s+2)(s+3)(s+4)} = \frac{12}{(s+2)} - \frac{24}{(s+3)} + \frac{12}{(s+4)}$$

$$C(s) = R(s) \frac{12}{(s+2)} - R(s) \frac{24}{(s+3)} + R(s) \frac{12}{(s+4)}$$

$$\dot{x}_1 = -2x_1 + 12r$$

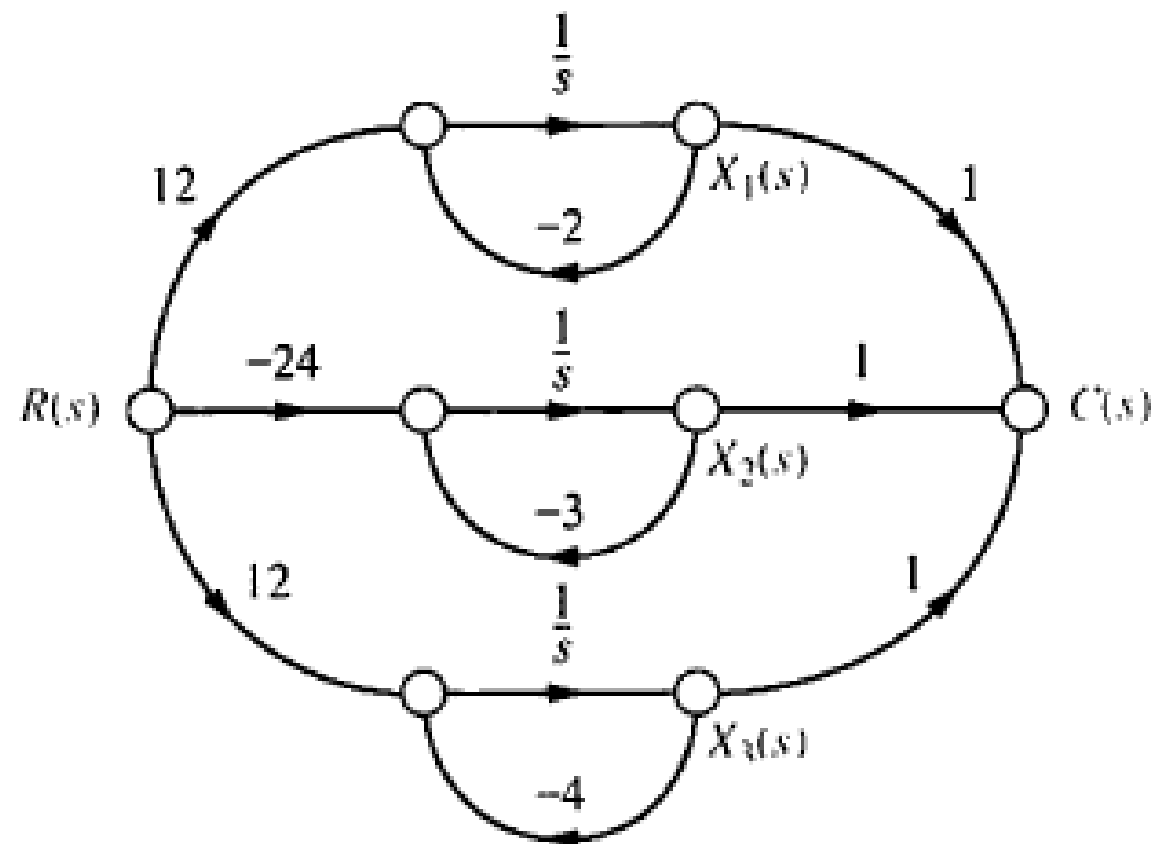
$$\dot{x}_2 = -3x_2 - 24r$$

$$\dot{x}_3 = -4x_3 + 12r$$

$$y = c(t) = x_1 + x_2 + x_3$$

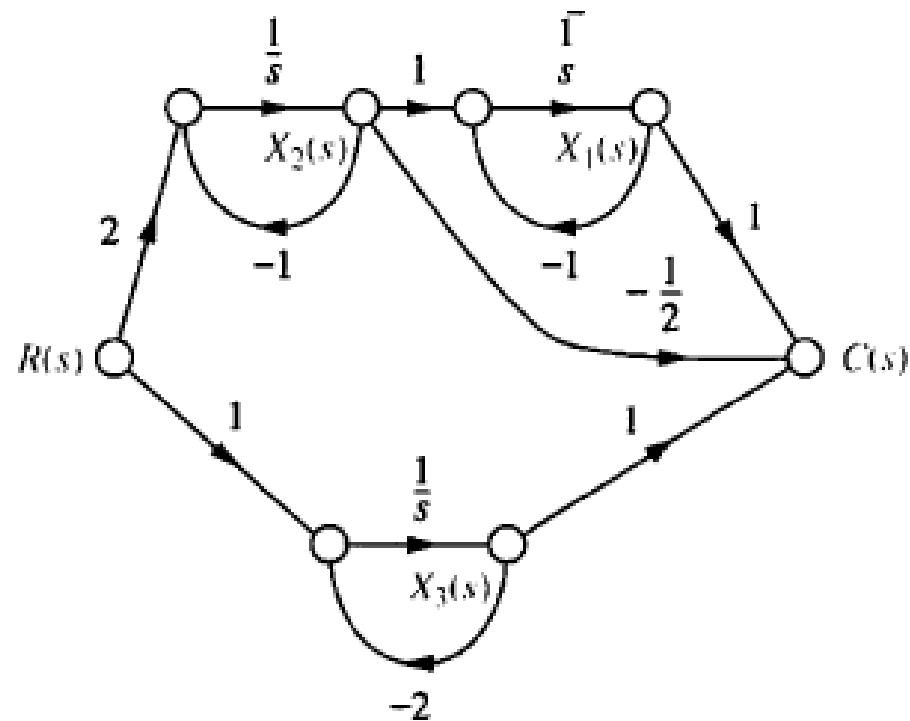
$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 12 \\ -24 \\ 12 \end{bmatrix} r$$

$$y = [1 \quad 1 \quad 1] \mathbf{x}$$



$$\frac{C(s)}{R(s)} = \frac{(s+3)}{(s+1)^2(s+2)}$$

$$\frac{C(s)}{R(s)} = \frac{2}{(s+1)^2} - \frac{1}{(s+1)} + \frac{1}{(s+2)}$$



$$\dot{x}_1 = -x_1 + x_2$$

$$\dot{x}_2 = -x_2 + 2r$$

$$\dot{x}_3 = -2x_3 + r$$

$$y = c(t) = x_1 - \frac{1}{2}x_2 + x_3$$

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & -\frac{1}{2} & 1 \end{bmatrix} \mathbf{x}$$

Controller canonical form

$$G(s) = \frac{C(s)}{R(s)} = \frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

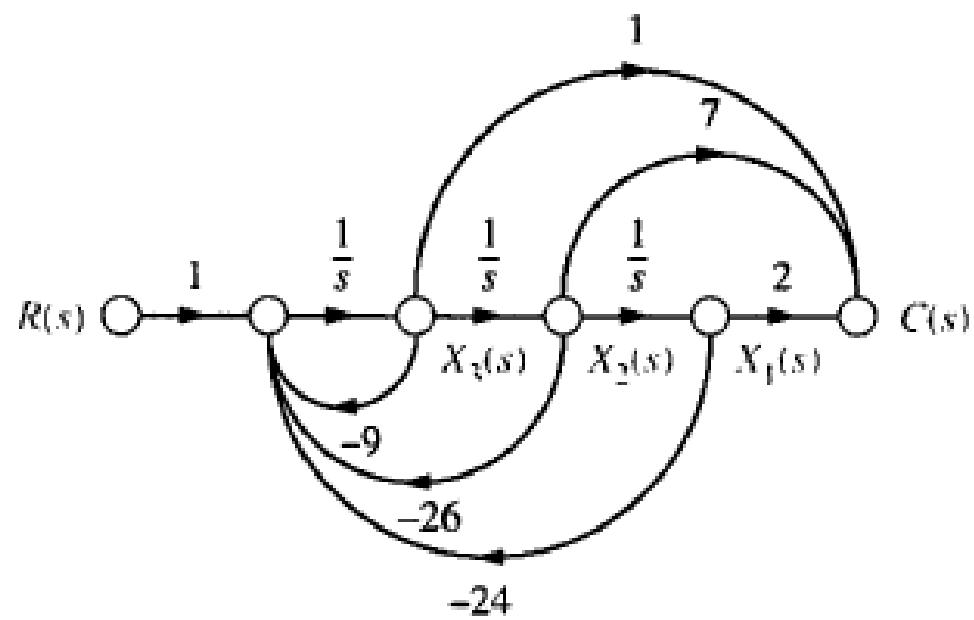
$$y = [2 \quad 7 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -9 & -26 & -24 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} r$$

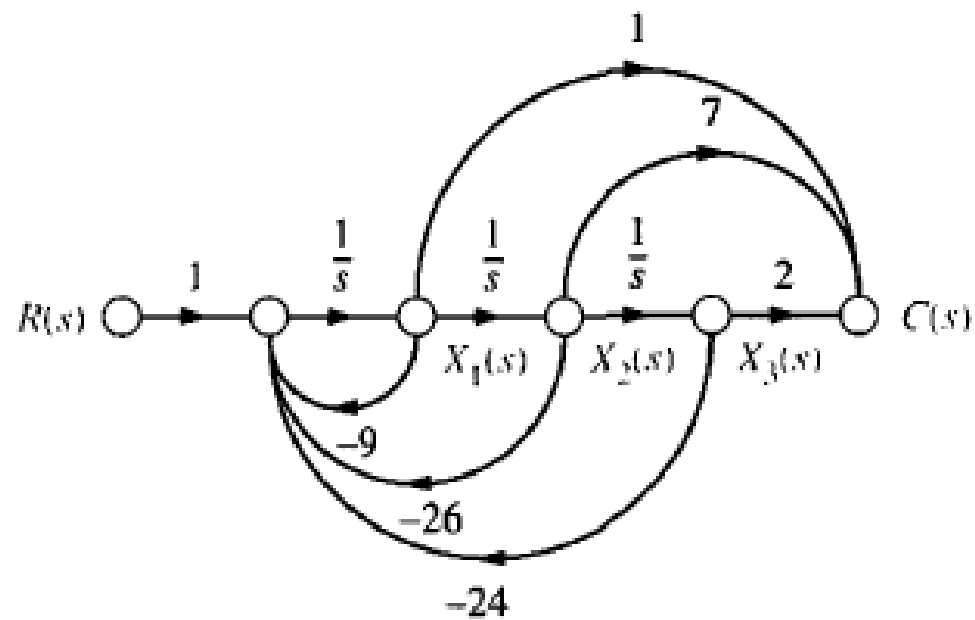
$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_2 \\ \dot{x}_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_3 \\ x_2 \\ x_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

$$y = [1 \quad 7 \quad 2] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y = [2 \quad 7 \quad 1] \begin{bmatrix} x_3 \\ x_2 \\ x_1 \end{bmatrix}$$



(a)



(b)

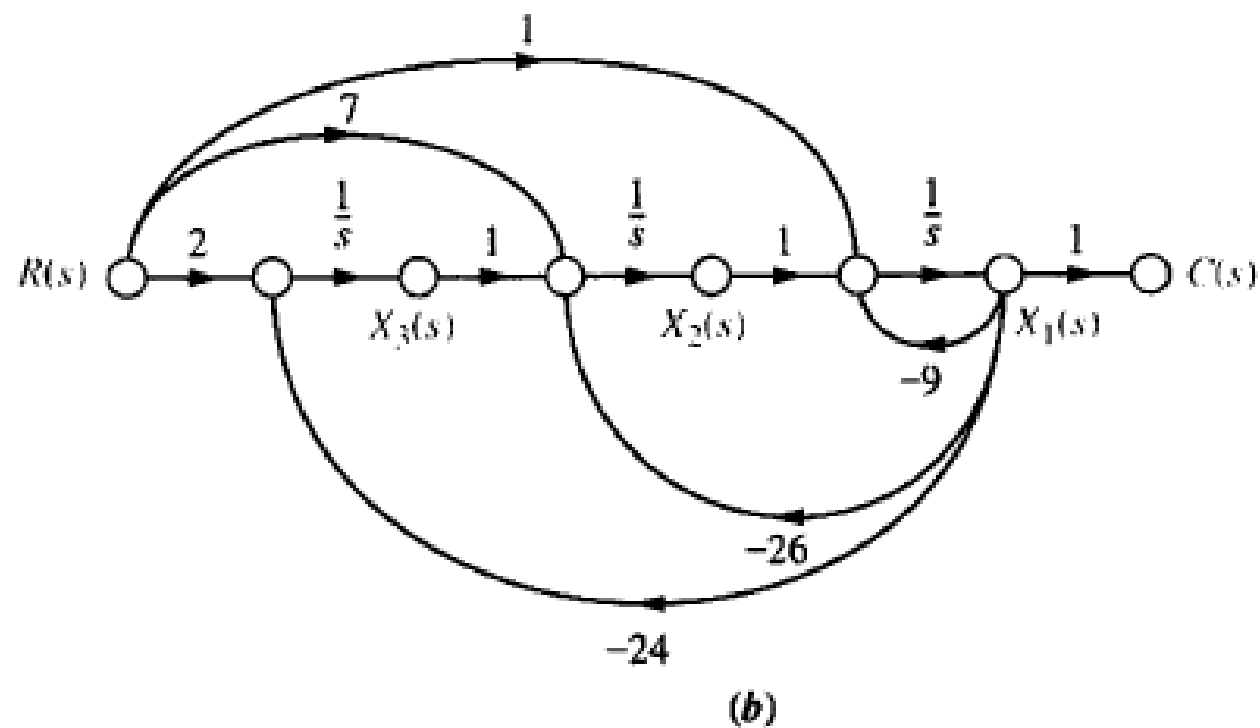
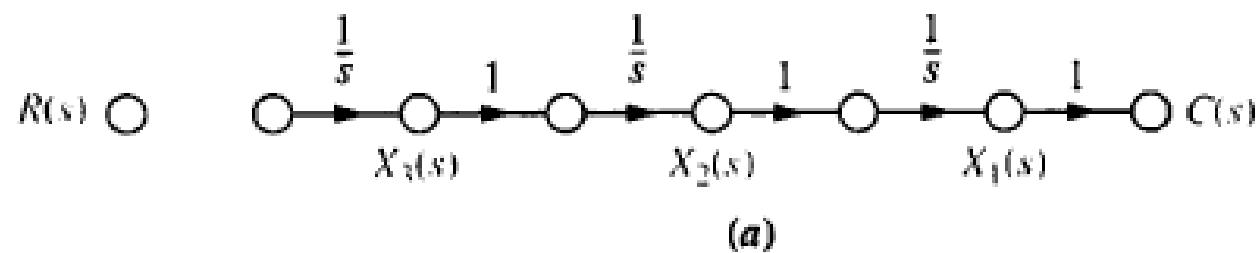
Observer canonical form

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{s} + \frac{7}{s^2} + \frac{2}{s^3}}{1 + \frac{9}{s} + \frac{26}{s^2} + \frac{24}{s^3}}$$

$$\left[\frac{1}{s} + \frac{7}{s^2} + \frac{2}{s^3} \right] R(s) = \left[1 + \frac{9}{s} + \frac{26}{s^2} + \frac{24}{s^3} \right] C(s)$$

$$C(s) = \frac{1}{s} [R(s) - 9C(s)] + \frac{1}{s^2} [7R(s) - 26C(s)] + \frac{1}{s^3} [2R(s) - 24C(s)]$$

$$C(s) = \frac{1}{s} \left[[R(s) - 9C(s)] + \frac{1}{s} \left([7R(s) - 26C(s)] + \frac{1}{s} [2R(s) - 24C(s)] \right) \right]$$



$$\dot{x}_1 = -9x_1 + x_2 + r$$

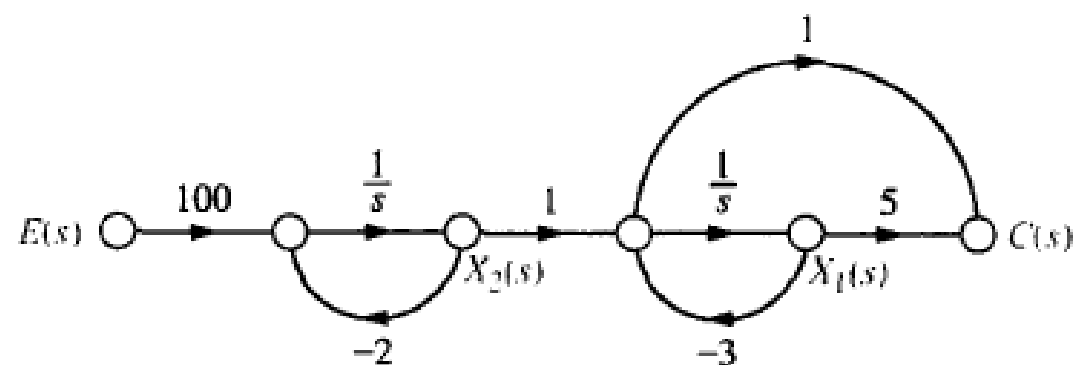
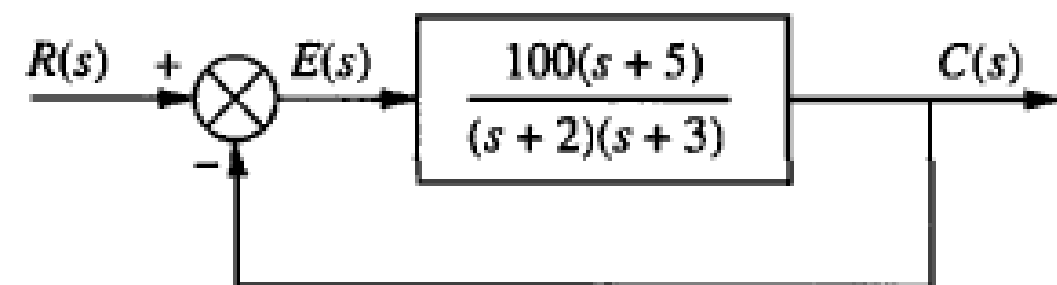
$$\dot{x}_2 = -26x_1 + x_3 + 7r$$

$$\dot{x}_3 = -24x_1 + 2r$$

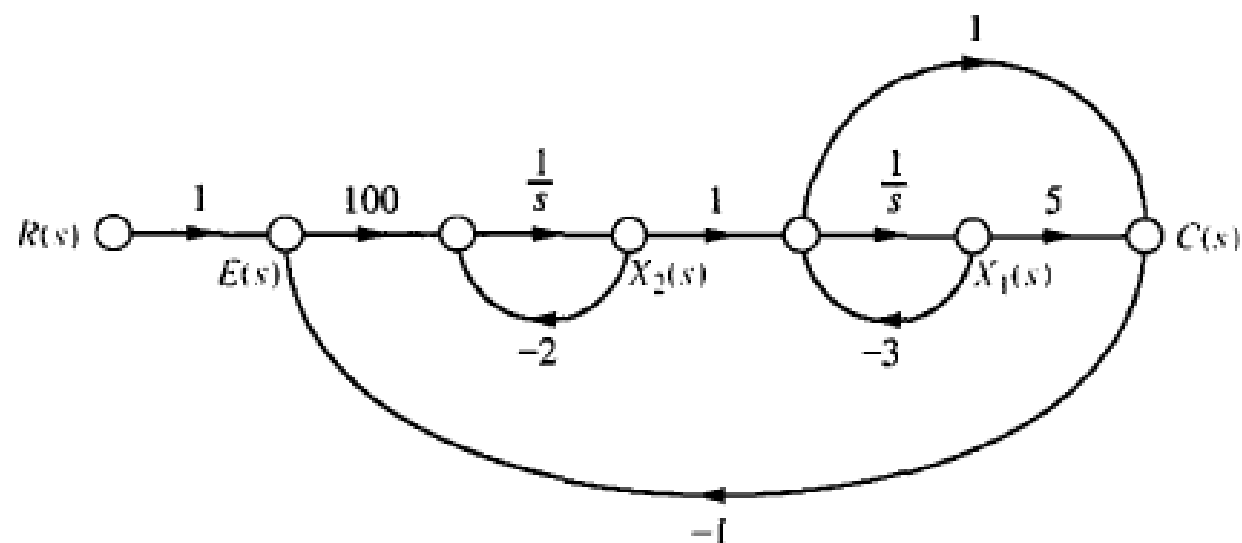
$$y = c(t) = x_1$$

$$\dot{\mathbf{x}} = \begin{bmatrix} -9 & 1 & 0 \\ -26 & 0 & 1 \\ -24 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix} r$$

$$y = [1 \quad 0 \quad 0] \mathbf{x}$$



(a)



(b)

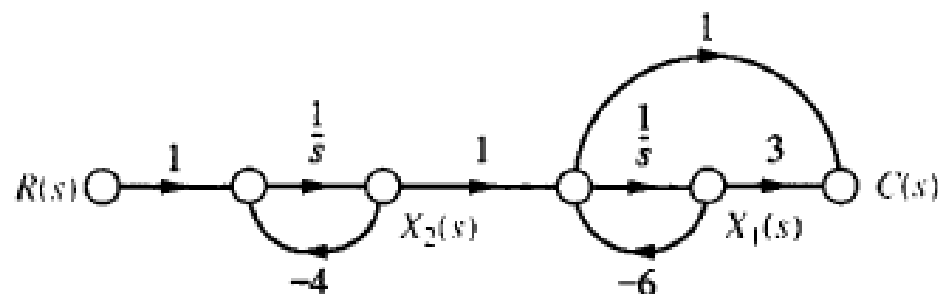
$$\dot{\mathbf{x}} = \begin{bmatrix} -3 & 1 \\ -200 & -102 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 100 \end{bmatrix} r$$

$$y = [2 \quad 1] \mathbf{x}$$

Form	Transfer function	Signal-flow diagram	State equations
Phase variable	$\frac{1}{(s^2 + 10s + 24)} * (s + 3)$		$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -24 & -10 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r$ $y = [3 \ 1] \mathbf{x}$
Parallel	$\frac{-1/2}{(s + 4)} + \frac{3/2}{s + 6}$		$\dot{\mathbf{x}} = \begin{bmatrix} -4 & 0 \\ 0 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -\frac{1}{2} \\ \frac{3}{2} \end{bmatrix} r$ $y = [1 \ 1] \mathbf{x}$
Cascade	$\frac{1}{(s + 4)} * \frac{(s + 3)}{(s + 6)}$		$\dot{\mathbf{x}} = \begin{bmatrix} -6 & 1 \\ 0 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r$ $y = [-3 \ 1] \mathbf{x}$

Cascade

$$\frac{1}{(s+4)} * \frac{(s+3)}{(s+6)}$$

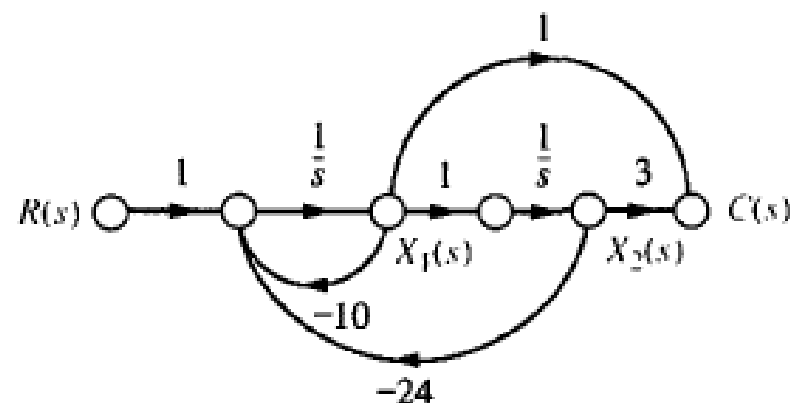


$$\dot{\mathbf{x}} = \begin{bmatrix} -6 & 1 \\ 0 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r$$

$$y = [-3 \ 1] \mathbf{x}$$

Controller canonical

$$\frac{1}{(s^2 + 10s + 24)} * (s + 3)$$

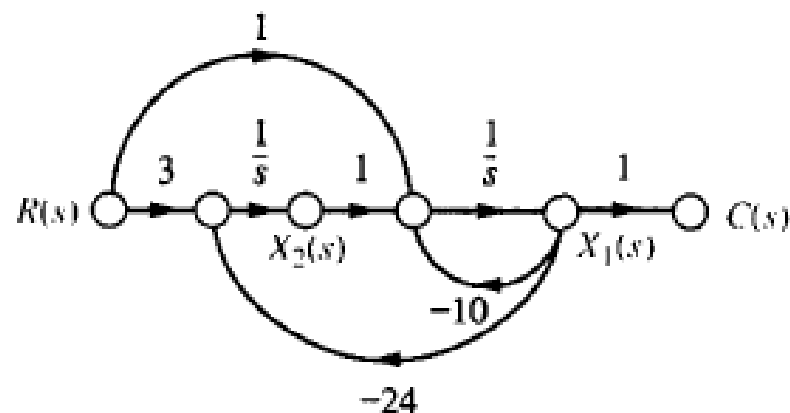


$$\dot{\mathbf{x}} = \begin{bmatrix} -10 & -24 \\ 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} r$$

$$y = [1 \ 3] \mathbf{x}$$

Observer canonical

$$\frac{\frac{1}{s} + \frac{3}{s^2}}{1 + \frac{10}{s} + \frac{24}{s^2}}$$



$$\dot{\mathbf{x}} = \begin{bmatrix} -10 & 1 \\ -24 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} r$$

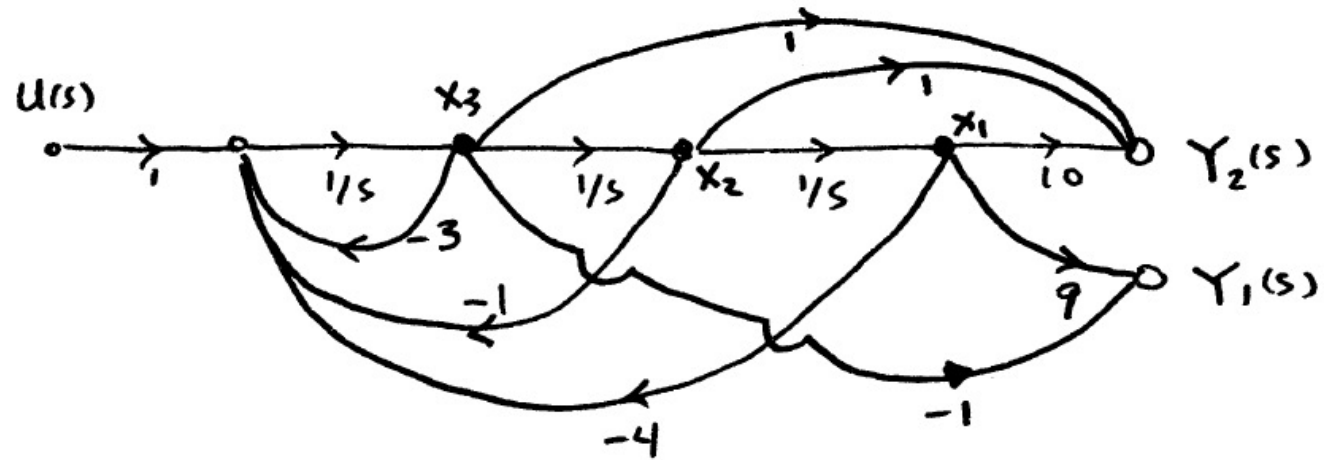
$$y = [1 \ 0] \mathbf{x}$$

FIGURE 5.31 State-space forms for $C(s)/R(s) = (s+3)/[(s+4)(s+6)]$. Note: $y = c(t)$

SIMO

$$T(s) = \frac{1}{s^3 + 3s^2 + s + 4} \begin{bmatrix} -s^2 + 9 \\ s^2 + s + 10 \end{bmatrix}$$

Solusi



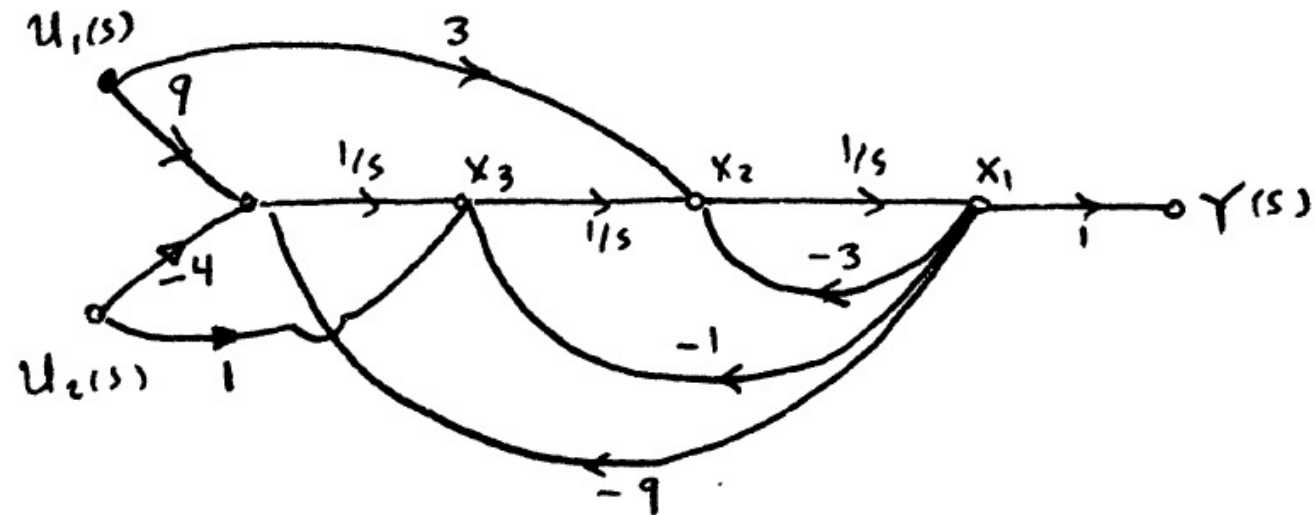
$$\dot{x} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -1 & -3 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$$

$$Y = \begin{pmatrix} 9 & 0 & -1 \\ 10 & 1 & 1 \end{pmatrix} x$$

MISO

$$T(s) = \frac{1}{s^3 + 3s^2 + s + 9} [3s^2 + 9 \quad s - 4]$$

Solusi



$$\dot{x} = \begin{bmatrix} -3 & 1 & 0 \\ -1 & 0 & 1 \\ -9 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 3 & 0 \\ 0 & 1 \\ 9 & -4 \end{bmatrix} u$$

$$y = [1 \quad 0 \quad 0] x$$

MIMO

$$\underline{T}(s) = \frac{1}{(s+1)(s+2)} \begin{bmatrix} 4s & s-3 \\ -6 & s+4 \end{bmatrix}$$

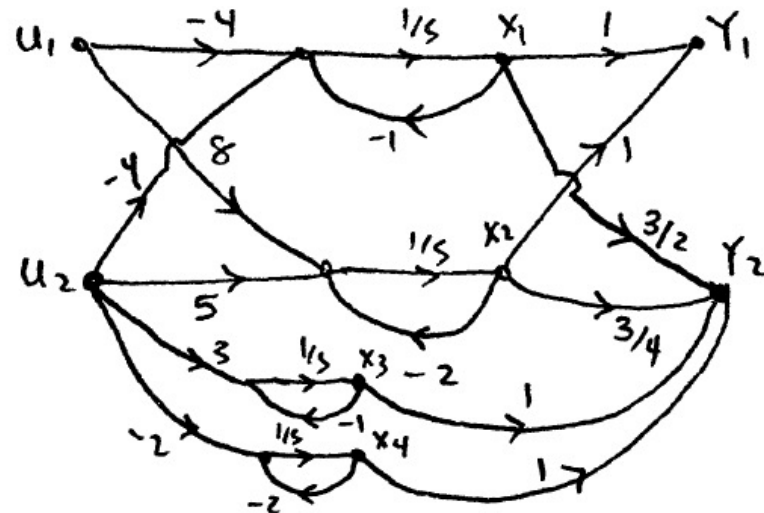
Solusi

$$= \begin{bmatrix} \frac{-4}{s+1} + \frac{8}{s+2} & \frac{-4}{s+1} + \frac{5}{s+2} \\ \frac{-6}{s+1} + \frac{6}{s+2} & \frac{3}{s+1} + \frac{-2}{s+2} \end{bmatrix}$$

$$X = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} X$$

$$+ \begin{bmatrix} -4 & -4 \\ 8 & 5 \\ 0 & 3 \\ 0 & -2 \end{bmatrix} U$$

$$Y = \begin{bmatrix} 1 & 1/4 & 0 & 0 \\ 3/2 & 3/4 & 1 & 1 \end{bmatrix} X$$

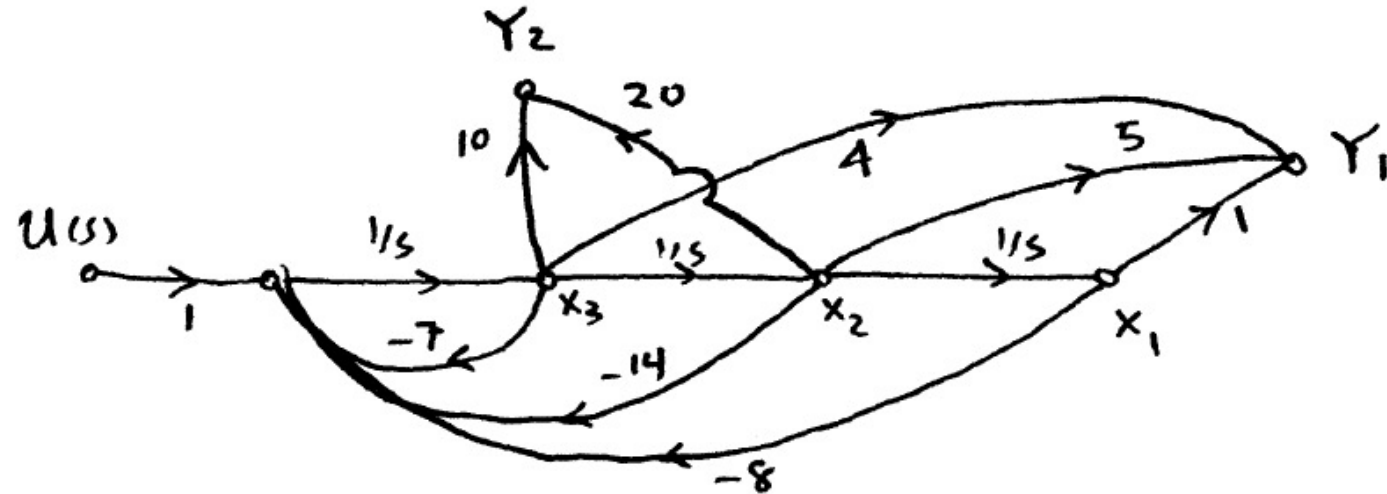


Latihan: No 1

$$T(s) = \frac{1}{(s+1)(s+2)(s+4)} \begin{bmatrix} (4s+1)(s+1) \\ 10s(s+2) \end{bmatrix}$$

Jawab

$$T(s) = \frac{1}{1 - \left(-\frac{7}{s} - \frac{14}{s^2} - \frac{8}{s^3}\right)} \left[\frac{4}{s} + \frac{5}{s^2} + \frac{1}{s^3} \right]$$



$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -14 & -7 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 20 & 10 \end{bmatrix} x$$

Latihan: No 2

$$T(s) = \left[\frac{(s-2)(s+2)}{s(s+1)(s+2)} \quad \frac{4(s-2)}{s(s+1)(s+2)} \quad \frac{(s+4)(s-1)}{s(s+1)(s+2)} \right]$$

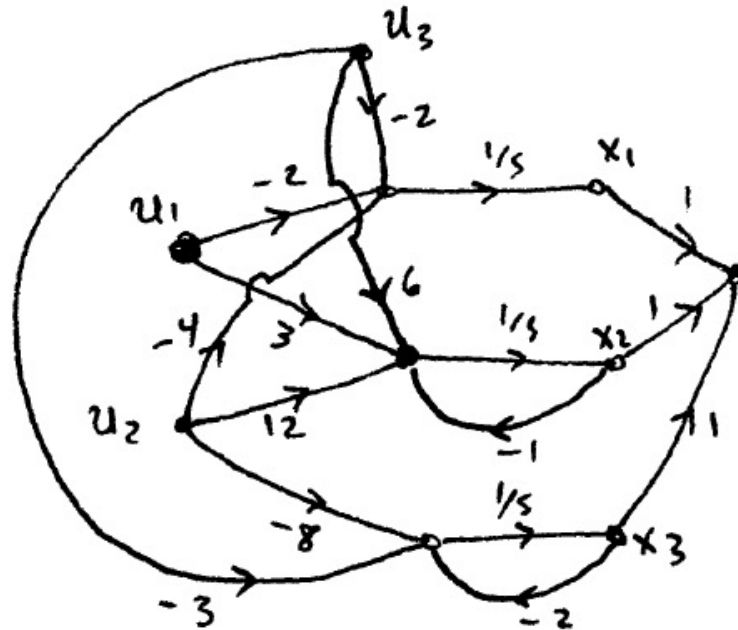
Jawab

$$= \begin{bmatrix} \frac{-2}{s} + \frac{3}{s+1} & \frac{-4}{s} + \frac{12}{s+1} + \frac{-8}{s+2} & \frac{-2}{s} + \frac{6}{s+1} + \frac{-3}{s+2} \end{bmatrix}$$

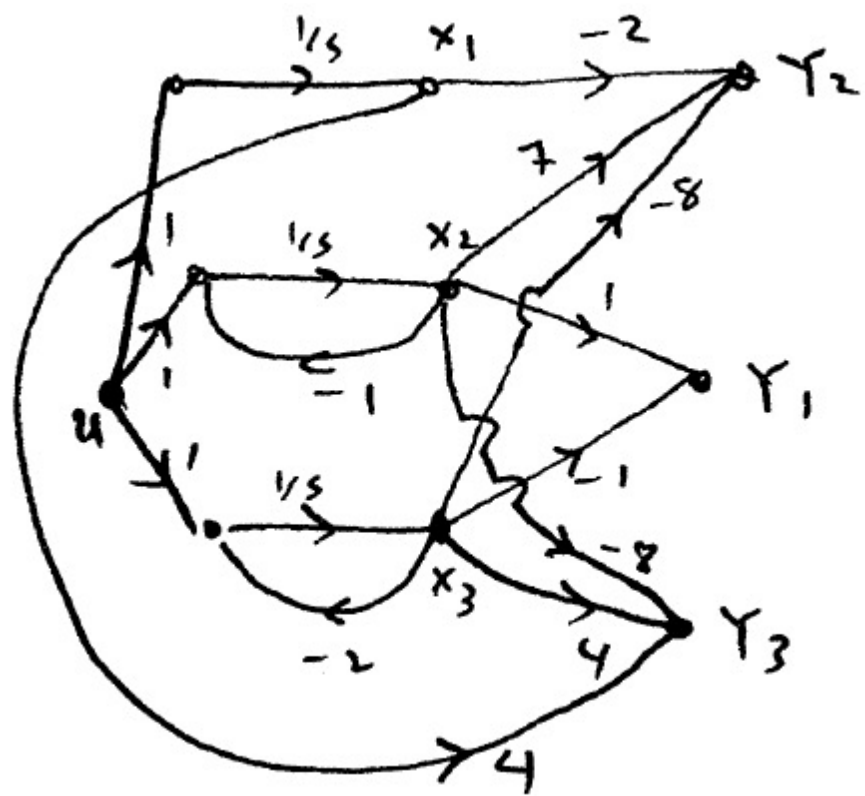
$$\dot{x} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & -2 \end{bmatrix} x +$$

$$+ \begin{bmatrix} -2 & -4 & -2 \\ 3 & 12 & 6 \\ 0 & -8 & -3 \end{bmatrix} u$$

$$y = [1 \quad 1 \quad 1] x$$



$$T = \frac{1}{s(s+1)(s+2)} \begin{bmatrix} s & -3s^2 - 4 \\ 8 \end{bmatrix} = \begin{bmatrix} \frac{1}{s+1} + \frac{-1}{s+2} \\ \frac{-2}{s} + \frac{7}{s+1} + \frac{-8}{s+2} \\ \frac{4}{s} + \frac{-8}{s+1} + \frac{4}{s+2} \end{bmatrix}$$



$$x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 & -1 \\ -2 & 7 & -8 \\ 4 & -8 & 4 \end{bmatrix} x$$

MIMO

To illustrate, consider a 2×2 (block) matrix

$$G(s) = \begin{bmatrix} G_1(s) & G_2(s) \\ G_3(s) & G_4(s) \end{bmatrix}$$

and assume that $G_i(s)$ has a state space realization of

$$G_i(s) = \left[\begin{array}{c|c} A_i & B_i \\ \hline C_i & D_i \end{array} \right], \quad i = 1, \dots, 4.$$

Note that $G_i(s)$ may itself be a MIMO transfer matrix.

Then a realization for $G(s)$ can be given by

$$G(s) = \left[\begin{array}{cccc|cc} A_1 & 0 & 0 & 0 & B_1 & 0 \\ 0 & A_2 & 0 & 0 & 0 & B_2 \\ 0 & 0 & A_3 & 0 & B_3 & 0 \\ 0 & 0 & 0 & A_4 & 0 & B_4 \\ \hline C_1 & C_2 & 0 & 0 & D_1 & D_2 \\ 0 & 0 & C_3 & C_4 & D_3 & D_4 \end{array} \right].$$

Let $G(s)$ be a $p \times m$ transfer matrix

$$G(s) = \frac{N(s)}{d(s)}$$

with $d(s)$ a scalar polynomial. For simplicity, we shall assume that $d(s)$ has only real and distinct roots $\lambda_i \neq \lambda_j$ if $i \neq j$ and

$$d(s) = (s - \lambda_1)(s - \lambda_2) \cdots (s - \lambda_r).$$

Then $G(s)$ has the following partial fractional expansion:

$$G(s) = D + \sum_{i=1}^r \frac{W_i}{s - \lambda_i}.$$

Suppose

$$\text{rank } W_i = k_i$$

and let $B_i \in \mathbb{R}^{k_i \times m}$ and $C_i \in \mathbb{R}^{p \times k_i}$ be two constant matrices such that

$$W_i = C_i B_i.$$

Then a realization for $G(s)$ is given by

$$G(s) = \left[\begin{array}{ccc|c} \lambda_1 I_{k_1} & & & B_1 \\ & \ddots & & \vdots \\ & & \lambda_r I_{k_r} & B_r \\ \hline C_1 & \cdots & C_r & D \end{array} \right].$$

Note that

$$\begin{aligned} G(s) &= \left[\begin{array}{cc|c} \lambda & 1 & b_1 \\ 0 & \lambda & b_2 \\ \hline c_1 & c_2 & 0 \end{array} \right] \\ &= \frac{c_1[b_2 + (s - \lambda)b_1]}{(s - \lambda)^2} + \frac{c_2b_2}{s - \lambda} \\ &= \frac{\mathbf{c}_1\mathbf{b}_2}{(s - \lambda)^2} + \frac{\mathbf{c}_1b_1 + c_2\mathbf{b}_2}{s - \lambda} \end{aligned}$$

A realization procedure:

- Let $G(s)$ be a $p \times q$ matrix and have the following partial fractional expansion:

$$G(s) = \frac{R_1}{(s - \lambda)^2} + \frac{R_2}{s - \lambda}$$

- Suppose $\text{rank}(R_1) = 1$ and write

$$R_1 = c_1 b_1, \quad c_1 \in \mathbb{R}^p, \quad b_1 \in \mathbb{R}^q$$

- Find c_2 and b_2 if possible such that

$$\mathbf{c}_1 b_1 + c_2 \mathbf{b}_2 = R_2$$

Otherwise find also matrices C_3 and B_3 such that

$$\mathbf{c}_1 b_1 + c_2 \mathbf{b}_2 + C_3 B_3 = R_2$$

and $[c_1 \ C_3]$ full column rank and $\begin{bmatrix} b_2 \\ B_3 \end{bmatrix}$ full row rank.

- if $\text{rank}(R_1) > 1$ then write

$$R_1 = c_1 b_1 + \tilde{c}_1 \tilde{b}_1 + \dots$$

and repeated the above process.

Consider a 3×3 transfer matrix:

$$G(s) = \begin{bmatrix} \frac{1}{(s+1)(s+2)} & \frac{2s+1}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \\ \frac{1}{(s+1)^2} & \frac{s^2+5s+3}{(s+1)^2} & \frac{s}{(s+1)^2} \\ \frac{1}{(s+1)^2(s+2)} & \frac{2s+1}{(s+1)^2(s+2)} & \frac{s}{(s+1)^2(s+2)} \end{bmatrix}.$$

$$\begin{aligned} G(s) &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \frac{1}{(s+1)^2} \overbrace{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}^{c_1} \overbrace{\begin{bmatrix} 1 & -1 & -1 \end{bmatrix}}^{b_2} \\ &+ \frac{1}{s+1} \left(\overbrace{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}^{c_1} \overbrace{\begin{bmatrix} 0 & 3 & 1 \end{bmatrix}}^{b_1} + \overbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}^{c_2} \overbrace{\begin{bmatrix} 1 & -1 & -1 \end{bmatrix}}^{b_2} \right) \\ &+ \frac{1}{s+1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} + \frac{1}{s+2} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -3 & -2 \end{bmatrix} \end{aligned}$$

So a 4-th order minimal state space realization is given by

$$G(s) = \left[\begin{array}{cccc|ccc} -1 & 1 & 0 & 0 & 0 & 3 & 1 \\ 0 & -1 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -2 & 1 & -3 & -2 \\ \hline 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \end{array} \right].$$

Let

$$\begin{aligned} G_3(s) &= \left[\begin{array}{ccc|c} \lambda & 1 & 0 & b_1 \\ 0 & \lambda & 1 & b_2 \\ 0 & 0 & \lambda & b_3 \\ \hline c_1 & c_2 & c_3 & 0 \end{array} \right] \\ &= \frac{c_1[b_3 + (s - \lambda)b_2 + (s - \lambda)^2b_1]}{(s - \lambda)^3} \\ &\quad + \frac{c_2[b_3 + (s - \lambda)b_2]}{(s - \lambda)^2} + \frac{c_3b_3}{s - \lambda} \\ &= \frac{\mathbf{c}_1\mathbf{b}_3}{(s - \lambda)^3} + \frac{\mathbf{c}_1b_2 + c_2\mathbf{b}_3}{(s - \lambda)^2} + \frac{\mathbf{c}_1b_1 + c_2b_2 + c_3\mathbf{b}_3}{s - \lambda} \end{aligned}$$

Example: Let

$$\begin{aligned}
 G(s) &= \begin{bmatrix} \frac{1}{(s+2)^3(s+5)} & \frac{1}{s+5} \\ \frac{1}{s+2} & 0 \end{bmatrix} \\
 &= \frac{1}{(s+2)^3} \overbrace{\begin{bmatrix} \frac{1}{3} \\ 0 \end{bmatrix}}^{c_1} \overbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}^{b_3} + \frac{1}{(s+2)^2} \overbrace{\begin{bmatrix} -\frac{1}{9} \\ 0 \end{bmatrix}}^{c_2} \overbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}^{b_3} \\
 &\quad + \frac{1}{s+2} \overbrace{\begin{bmatrix} \frac{1}{27} \\ 1 \end{bmatrix}}^{c_3} \overbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}^{b_3} + \frac{1}{s+5} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} -\frac{1}{27} & 1 \end{bmatrix}
 \end{aligned}$$

Take $b_1 = 0$ and $b_2 = 0$, we get

$$G(s) = \left[\begin{array}{cccc|cc} -2 & 1 & 0 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & -5 & -\frac{1}{27} & 1 \\ \hline \frac{1}{3} & -\frac{1}{9} & \frac{1}{27} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array} \right]$$

Example: Let

$$\begin{aligned}
 G(s) = & \frac{\overbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}^{c_1} \overbrace{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}^{b_3}}{(s+p)^3} + \frac{\overbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}^{\tilde{c}_1} \overbrace{\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}}^{\tilde{b}_3}}{(s+p)^3} \\
 & + \frac{\overbrace{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}^{c_2} \overbrace{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}^{b_3}}{(s+p)^2} + \frac{\overbrace{\begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}}^{2c_1+3\tilde{c}_1} \overbrace{\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}}^{\frac{1}{2}b_2 \text{ or } \frac{1}{3}\tilde{c}_1}}{(s+p)^2} \\
 & + \frac{\overbrace{\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}}^{-c_3 \text{ or } -\tilde{c}_3} \overbrace{\begin{bmatrix} -1 & -1 & 0 \end{bmatrix}}^{-b_3-\tilde{b}_3}}{s+p}
 \end{aligned}$$

Hence

$$G(s) = \left[\begin{array}{cccccc|ccc} -p & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -p & 1 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & -p & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -p & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -p & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & -p & 0 & 1 & 0 \\ \hline 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$